GLM Basics Guide

Overview

GLM (General Lake Model) is a 1D hydrodynamic model that simulates vertical temperature and mixing dynamics in lakes. It reads configuration from a namelist file and produces NetCDF output.

Running GLM
```bash
cd /root
glm
```

GLM reads `glm3.nml` in the current directory and produces output in `output/output.nc`.

Input File Structure

File	Description
`glm3.nml`	Main configuration file (Fortran namelist format)
`bcs/*.csv`	Boundary condition files (meteorology, inflows, outflows)

Configuration File Format

`glm3.nml` uses Fortran namelist format with multiple sections:
```fortran
&glm_setup
sim_name = 'LakeName'
max_layers = 500
```

```
/
&light
Kw = 0.3
/
&mixing
coef_mix_hyp = 0.5
/
&meteorology
meteo_fl = 'bcs/meteo.csv'
wind_factor = 1
lw_factor = 1
ch = 0.0013
/
&inflow
inflow_fl = 'bcs/inflow1.csv','bcs/inflow2.csv'
/
&outflow
outflow_fl = 'bcs/outflow.csv'
/
```

Modifying Parameters with Python

```python
import re

def modify_nml(nml_path, params):
with open(nml_path, 'r') as f:
content = f.read()
for param, value in params.items():
pattern = rf"({param}\s*=\s*)[\d\.\-e]+"
replacement = rf"\g<1>{value}"
content = re.sub(pattern, replacement, content)
with open(nml_path, 'w') as f:
f.write(content)

# Example usage
modify_nml('glm3.nml', {'Kw': 0.25, 'wind_factor': 0.9})
```

Common Issues

Issue	Cause	Solution
GLM fails to start	Missing input files	Check bcs/ directory
No output generated	Invalid nml syntax	Check namelist format
Simulation crashes	Unrealistic parameters	Use values within valid ranges

Best Practices

- Always backup `glm3.nml` before modifying
- Run GLM after each parameter change to verify it works
- Check `output/` directory for results after each run